

Table joins

Cartesian (cross join)

	employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id	c
1	100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90	
2	101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.00	NULL	100	90	
3	102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90	
4	103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.00	NULL	102	60	

```
select *  
from employees, departments
```

2889 rows in result (107 employees * 27 departments)

Cartesian with condition

	employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id	department_id
1	100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90	90
2	101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.00	NULL	100	90	90
3	102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90	90
4	103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.00	NULL	102	60	60
5	104	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_PROG	6000.00	NULL	103	60	60

```
select *  
from employees e, departments d  
where e.department_id=d.department_id
```

	employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id	department_id
1	100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90	90
2	101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.00	NULL	100	90	90
3	102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90	90
4	103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.00	NULL	102	60	60
5	104	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_PROG	6000.00	NULL	103	60	60
6	105	Daniel	Mathews	DMATHAMS	1.590.555.0105	2015-06-25	IT_PROG	4000.00	NULL	102	60	60

Inner join

```
select *
      from employees e join departments d
      on e.department_id=d.department_id
```

Result: 106 rows

Inner join with not equal operator

	last_name	salary	min_salary	max_salary
1	James	9000.00	4200	9000
2	Miller	6000.00	4200	9000
3	Williams	4800.00	4200	9000
4	Jackson	4800.00	4200	9000
5	Nguyen	4200.00	4200	9000
6	Faviet	9000.00	4200	9000
7	Chen	8200.00	4200	9000
8	Sciarra	7700.00	4200	9000

```
select
e.last_name,
e.salary,
j.min_salary,
j.max_salary
  from employees e join jobs j  on e.salary between j.min_salary and j.max_salary
```

Result: 643 rows

Right join

	department_id	department_id	department_name
103	100	100	Finance
104	100	100	Finance
105	110	110	Accounting
106	110	110	Accounting
107	NULL	120	Treasury
108	NULL	130	Corporate Tax
109	NULL	140	Control And Credit
110	NULL	150	Shareholder Ser...
111	NULL	160	Benefits

```
select
e.department_id,
d.department_id,
d.department_name
  from employees e right join departments d
    on e.department_id=d.department_id
```

Result: 122 rows

Left join

	last_name	department_id	department_id	department_name
73	Bates	80	80	Sales
74	Kumar	80	80	Sales
75	Abel	80	80	Sales
76	Hutton	80	80	Sales
77	Taylor	80	80	Sales
78	Livingston	80	80	Sales
79	Grant	NULL	NULL	NULL
80	Johnson	80	80	Sales
81	Taylor	50	50	Shipping

```
select
e.last_name,
e.department_id,
d.department_id,
d.department_name
  from employees e left join departments d
    on e.department_id=d.department_id
```

Result: 107 rows

Full join

	last_name	department_id	department_id	department_name
1	Grant	NULL	NULL	NULL
2	NULL	NULL	120	Treasury
3	NULL	NULL	130	Corporate Tax
4	NULL	NULL	140	Control And Credit
5	NULL	NULL	150	Shareholder Services
6	NULL	NULL	160	Benefits
7	NULL	NULL	170	Manufacturing
8	NULL	NULL	180	Construction
9	NULL	NULL	190	Contracting
10	NULL	NULL	200	Operations
11	NULL	NULL	210	IT Support
12	NULL	NULL	220	NOC
13	NULL	NULL	230	IT Helpdesk
14	NULL	NULL	240	Government Sales
15	NULL	NULL	250	Retail Sales
16	NULL	NULL	260	Recruiting
17	NULL	NULL	270	Payroll
18	Whalen	10	10	Administration

```
select
e.last_name,
e.department_id,
d.department_id,
d.department_name
  from employees e full join departments d
    on e.department_id=d.department_id
 order by e.department_id
Result: 123 rows
```


Self join

	employee_id	last_name	manager_id	employee_id	last_name	manager_id
1	100	King	NULL	101	Yang	100
2	100	King	NULL	102	Garcia	100
3	100	King	NULL	114	Li	100
4	100	King	NULL	120	Weiss	100
5	100	King	NULL	121	Fripp	100
6	100	King	NULL	122	Kaufling	100
7	100	King	NULL	123	Vollman	100
8	100	King	NULL	124	Mourgos	100
9	100	King	NULL	145	Singh	100
10	100	King	NULL	146	Partners	100
11	100	King	NULL	147	Errazuriz	100
12	100	King	NULL	148	Cambraut	100
13	100	King	NULL	149	Zlotkey	100
14	100	King	NULL	201	Martinez	100
15	101	Yang	100	200	Whalen	101
16	101	Yang	100	203	Jacobs	101

```
select
e.employee_id,
e.last_name,
e.manager_id,
b.employee_id,
b.last_name,
b.manager_id
  from employees e left join employees b
    on e.employee_id=b.manager_id
```

Result: 195 rows

Anti join

	last_name	department_id	department_id	department_name
1	Abel	80	10	Administration
2	Abel	80	20	Marketing
3	Abel	80	30	Purchasing
4	Abel	80	40	Human Resources
5	Abel	80	50	Shipping
6	Abel	80	60	IT
7	Abel	80	70	Public Relations
8	Abel	80	90	Executive
9	Abel	80	100	Finance
10	Abel	80	110	Accounting
11	Abel	80	120	Treasury

```
select
e.last_name,
e.department_id,
d.department_id,
d.department_name
  from employees e join departments d
    on e.department_id != d.department_id
 order by 1
```

Result: 2756 rows

Join with
subquery

	employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id	department_name	department_id
1	100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90	Executive	90
2	101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.00	NULL	100	90	Executive	90
3	102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90	Executive	90
4	103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.00	NULL	102	60	IT	60
5	104	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_PROG	6000.00	NULL	103	60	IT	60
6	105	David	Williams	DWILLIAMS	1.590.555.0105	2015-06-25	IT_PROG	4800.00	NULL	103	60	IT	60
7	106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_PROG	4800.00	NULL	103	60	IT	60

```
select
```

```
*
```

```
from employees e
```

```
join
```

```
(select department_name, department_id from departments) d  
on e.department_id = d.department_id
```

CROSS APPLY (inner)

```
CREATE FUNCTION depart(@dept_id int)
RETURNS TABLE
AS
RETURN
    SELECT
        department_id,
        department_name
    from departments
    where department_id = @dept_id

select
    count(*) liczba_wierszy
from employees e
    cross apply
        depart(e.department_id)
```

	liczba_wierszy
1	106

OUTER APPLY (outer)

```
CREATE FUNCTION depart(@dept_id int)
RETURNS TABLE
AS
RETURN
    SELECT
        department_id,
        department_name
    from departments
    where department_id = @dept_id

select
    count(*) liczba_wierszy
from employees e
    outer apply
        depart(e.department_id)
```

	liczba_wierszy
1	107

Union

Results		Messages	
	a	b	
1	1	2	
2	1	3	

```
select 1 as a, 2 as b
union
select 1 , 2
union
select 1 , 3
```

Union all

Results			
	a	b	
1	1	2	
2	1	2	
3	1	3	

```
select 1 as a, 2 as b
union all
select 1 , 2
union all
select 1 , 3
```

Intersect

Results		
	a	b
1	1	2

```
select 1 as a, 2 as b  
intersect  
select 1, 2
```


Except

Results		Message
	a	b
1	1	3

```
select 1 as a, 3 as b
except
select 1 , 4
except
select 1 , 4
```